

St. Philomena's College (Autonomous), Mysuru
I Semester - BCA – Final Examination : Nov / Dec - 2024
Subject: COMPUTER APPLICATIONS
Title : Problem Solving Using C++

Time: 3 Hours

Max. Marks: 80

PART - A**I Answer any TEN questions:-****10x2=20**

- 1. What is the difference between cout and cin in C++ ?
- 2. Define variable in C++ with an example.
- 3. Define Abstraction.
- 4. Define Functions in C++ with Syntax.
- 5. Mention any four Math.h functions.
- 6. Write the syntax for do-while loop.
- 7. Define class and object.
- 8. What is the purpose of a constructor in C++ ?
- 9. What is the difference between private and public access specifiers.
- 10. Define operator overloading.
- 11. Write the syntax for multi-level inheritance.
- 12. Define Polymorphism.

PART - B**II Answer any SIX questions :****6x10=60**

- *• 13. a) Explain the characteristics of OOP's. *void int Boolean* 5
b) Explain Primary data types in C++ with example. 5
- *• 14. List the operators supported in C++. Explain any three operators with example programs. 10
- *• 15. Explain Decision Making Statements in C++ with example programs. 10
- 16. What is function overloading? What a C++ Program to illustrate function overloading. 10
- 17. Explain any 5 string manipulation functions in C++ with example programs. 10
- 18. a) What is an array? How to declare and initialize one dimensional array in C++ .Illustrate with example program. 10
- 19. Explain different types of inheritance in C++ with program. 10
- *• 20. Explain C++ file handling functions with syntax. 10
- 21. Explain Program Development Life Cycle. 10

Null


St. Philomena's College (Autonomous) Mysore
I Semester BCA – Final Examination : November / December - 2024
Subject : COMPUTER APPLICATIONS
Title : Digital Computer Organisation

Time : 3 Hours

Max Marks : 80

PART - A

Answer any TEN of the following :

10×2=20

- 1. Define Algorithm.
- 2. Define software? Mention the types.
- 3. Expand EEPROM and LCD.
- 4. $1011_{(2)} \times 1101_{(2)} = ?_{(2)}$
- 5. What are weighted codes?
- 6. Convert 10101 to gray code.
- 7. Define Boolean algebra? Mention one basic law.
- 8. State duality theorem with an example.
- 9. Write the truth table of EX-OR gate.
- 10. What is half adder?
- 11. Mention the applications of flip flop.
- 12. What is shift register?

PART - B

Answer any SIX of the following :

6×10=60

- 13. Explain the features of various generations of computer.
- 14. a) Differences between Primary memory and secondary memory.
b) Write a note on scanners.
- 15. a) Discuss different types of printers.
b) Explain ALL and HLL.
- 16. a) Convert $(10111001)_2$ to octal, Hexadecimal and decimal number systems.
b) Perform BCD addition:
i) $128_{(10)} + 12_{(10)}$ ii) $19_{(10)} + 20_{(10)}$
- 17. a) Subtract $127_{(10)}$ from $98_{(10)}$ using 1's and 2's method.
b) Write a note on Non-Weighted Codes.

PTO

18. a) State and prove Demorgan's 1st theorem. 05
b) Solve the following Boolean expression using Kmap 05
$$A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C$$
19. Construct basic gates using universal gates. 10
20. a) Explain full adder. 05
b) Explain SR and clocked SR flip flop with neat diagram. 05
21. a) Explain 3-8 line decoders. 05
b) Write a short note on ripple counter. 05

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St. Philomena's College (Autonomous) Mysore
I Semester BCA – Final Examination : November / December - 2024
Subject : COMPUTER APPLICATIONS
Title : Mathematical and Statistical Computing using R

Time : 3 Hours

Max Marks : 80

PART - A

Answer any TEN of the following :

10×2=20

1. Construct Truth table for the compound statement $p \rightarrow (q \rightarrow r)$.
2. What is Tautology?
3. If $A = \{1,2,3,4,5\}$ and $B = \{4,5,6\}$. Find $A - B$ and write the venn diagram.
4. Mention the properties of Relations.
5. What is an equivalence Relation? Give an example.
6. Define one – to – one and onto function.
7. Determine mean for the first 5 Prime numbers.
8. What is Frequency Distribution? Mention its types.
9. What is median, mode? Write the formula to find the same.
10. Define standard Deviation.
11. What is co-efficient of variation? Write its formula.
12. Write the difference between correlation and Regression.

PART – B

Answer any SIX of the following :

6×10=60

13. Explain the following connectives with appropriate examples. 10
a) Conjunction b) Disjunction c) Negation
14. Mention different set operations. Explain (i) Union (ii) Intersection with examples. 10
15. a) What is Relation? Explain and find the given relation is Reflexive or not $A = \{1,2,3,4\}$ 05
 $R_1 = \{(1,1)(1,2)(2,1)(2,2)(3,4)(4,1)(4,4)\}$
 $R_2 = \{(1,1)(1,2)(1,4)(2,1)(2,2)(3,3)(4,1)(4,4)\}$
 $R_3 = \{(1,1)(1,2)(1,3)(1,4)(2,2)(2,3)(2,4)(3,3)(3,4)(4,4)\}$
 $R_4 = \{(1,1)(1,2)(2,1)\}$
b) Check whether $(p \vee q) \wedge (\sim p \wedge \sim q)$ is a tautology or contradiction. 05

✓ 16. a) What is measures of central tendency?

02

b) Calculate the Mean and Median for the following data.

08

x	10	15	20	25	30
f	1	2	3	4	5

17. a) Mention the differences between Grouped and ungrouped data.

05

b) What is Discrete series and continuous series. Explain with an example.

05

✓ 18. a) Find the mode for the following series.

08

CI	0-10	10-20	20-30	30-40	40-50
Frequency	7	10	15	8	9

b) Calculate the median for the first ten even numbers.

02

✓ 19. Write the steps and calculate the standard deviation for the following data using the actual mean method.

10

Size(X)	5	10	15	20	25	30
Frequency(f)	6	7	3	2	1	1

20. Explain and mention properties of Regression Co-efficients.

10

✓ 21. What is Correlation? Explain its types. Calculate Rank Correlation co-efficient between 'x' and 'y' variables.

10

x	10	20	35	14	18	21	16
y	15	25	18	19	20	20	27
